

Report for ForestQuery into Global Deforestation, 1990 to 2016

ForestQuery is on a mission to combat deforestation around the world and to raise awareness about this topic and its impact on the environment. The data analysis team at ForestQuery has obtained data from the World Bank that includes forest area and total land area by country and year from 1990 to 2016, as well as a table of countries and the regions to which they belong.

The data analysis team has used SQL to bring these tables together and to query them in an effort to find areas of concern as well as areas that present an opportunity to learn from successes.

1. GLOBAL SITUATION

According to the World Bank, the total forest area of the world was 41282694.9 in 1990. As of 2016, the most recent year for which data was available, that number had fallen to 39958245.9, a loss of 1324449, or 3.2%.

The forest area lost over this time period is slightly more than the entire land area of Peru listed for the year 2016 (which is 1279999.9891).

2. REGIONAL OUTLOOK

In 2016, the percent of the total land area of the world designated as forest was 31.38%. The region with the highest relative forestation was Latin America & Carribean, with 46.16%, and the region with the lowest relative forestation was Middle East & North Africa, with _2.07% forestation.

In 1990, the percent of the total land area of the world designated as forest was 32.42%. The region with the highest relative forestation was_Latin America & Carribean, with 51.03%, and the region with the lowest relative forestation was Middle East & North Africa , with 1.78%% forestation.

Table 2.1: Percent Forest Area by Region, 1990 & 2016:

Region	1990 Forest Percentage	2016 Forest Percentage
Latin America & Caribbean	51.03	46.16
Europe & Central Asia	37.28	38.04
North America	35.65	36.04
Sub-Saharan Africa	30.67	28.79
East Asia & Pacific	25.78	26.36
South Asia	16.51	17.51
Middle East & North Africa	1.78	2.07

The only regions of the world that decreased in percent forest area from 1990 to 2016 were Latin America & Caribbean (dropped from 51.03% to 46.16%) and Sub-Saharan Africa (30.67 % to 28.79%). All other regions actually increased in forest area over this time period. However, the drop in forest area in the two aforementioned regions was so large, the percent forest area of the world decreased over this time period from 32.42% to 31.8%.

3. COUNTRY-LEVEL DETAIL

A. SUCCESS STORIES

There is one particularly bright spot in the data at the country level, China. This country actually increased in forest area from 1990 to 2016 by 527229.06. It would be interesting to study what has changed in this country over this time to drive this figure in the data higher. The country with the next largest increase in forest area from 1990 to 2016 was the United States, but it only saw an increase of 79200.00 sq km, much lower than the figure for China .

China and the United States are of course very large countries in total land area, so when we look at the largest *percent* change in forest area from 1990 to 2016, we aren't surprised to find a much smaller country listed at the top. Iceland's forest area by 213.66% from 1990 to 2016.

B. LARGEST CONCERNS

Which countries are seeing deforestation to the largest degree? We can answer this question in two ways. First, we can look at the absolute square kilometer decrease in forest area from 1990 to 2016. The following 3 countries had the largest decrease in forest area over the time period under consideration:

Table 3.1: Top 5 Amount Decrease in Forest Area by Country, 1990 & 2016:

Country	Region	Absolute Forest Area Change
Brazil	Latin America & Caribbean	541510
Indonesia	East Asia & Pacific	282193.98
Myanmar	East Asia & Pacific	107234
Nigeria	Sub-Saharan Africa	106506
Tanzania	Sub-Saharan Africa	102320

The second way to consider which countries are of concern is to analyze the data by percent decrease.

Table 3.2: Top 5 Percent Decrease in Forest Area by Country, 1990 & 2016:

Country	Region	Pct Forest Area Change
Togo	Sub-Saharan Africa	75.45
Nigeria	Sub-Saharan Africa	61.8
Uganda	Sub-Saharan Africa	59.13
Mauritania	Sub-Saharan Africa	46.75
Honduras	Latin America & Caribbean	45.03

When we consider countries that decreased in forest area percentage the most between 1990 and 2016, we find that four of the top 5 countries on the list are in the region of Sub-Saharan Africa.. The countries areTango, Nigeria, Uganda, and Mauritania . The 5th country on the list is Honduras,, which is in the Latin America & Caribbean region.

From the above analysis, we see that Nigeria is the only country that ranks in the top 5 both in terms of absolute square kilometer decrease in forest as well as percent decrease in forest area from 1990 to 2016. Therefore, this country has a significant opportunity ahead to stop the decline and hopefully spearhead remedial efforts.

C. QUARTILES

Table 3.3: Count of Countries Grouped by Forestation Percent Quartiles, 2016:

Quartile	Number of Countries
1	85
2	72
3	38
4	9

The largest number of countries in 2016 were found in the first quartile.

There were 9 countries in the top quartile in 2016. These are countries with a very high percentage of their land area designated as forest. The following is a list of countries and their respective forest land, denoted as a percentage.

Table 3.4: Top Quartile Countries, 2016:

Country	Region	Pct Designated as Forest
Suriname	Latin America & Caribbean	98.26
Micronesia	East Asia & Pacific	91.86
Gabon	Sub-Saharan Africa	90.04
Seychelles	Sub-Saharan Africa	88.41
Palau	East Asia & Pacific	87.61

American Samoa	East Asia & Pacific	87.5
Guyana	Latin America & Caribbean	83.9
Lao PDR	East Asia & Pacific	82.11
Solomon Islands	East Asia & Pacific	77.86

There are 94 countries with percent forestation higher than the United States in 2016

4. RECOMMENDATIONS

Write out a set of recommendations as an analyst on the ForestQuery team.

- *What have you learned from the World Bank data?*

Forest is slowly disappearing. The data analysis shows reduction of the global forest between 1990 and 2016. The most impacted region is Sub-Saharan Africa. For example, Togo reduced the forest by 75.45%. Even if we look at the count of countries grouped by forestation percent, we can see that 85 countries are in the first quartile with forestation between 0 -25% and 72 countries belong to the second quartile with forestation at level between 25 and 50%.

- *Which countries should we focus on over others?*

Which countries should we focus on over others?

4 out of 5 countries with top percent decrease are in Sub-Saharan Africa. Togo, lost in the analysed period (1990 -2016) over 75% of the forest. Other countries we should focus on are Nigeria (61.80%), Uganda (59.13%) and Mauritania (46.75%). People need to understand the way of life is having an unsustainable ecological footprint. There are several things which could be done in order to decrease this footprint. Firstly, consume less. Decrease in production of goods can slow down the deforestation trend. Secondly, avoid products containing Palm oil which causes big deforestation mainly in Asia. Thirdly, shop sustainably certified products.

5. APPENDIX: SQL Queries Used

```
-- =====
-- ForestQuery / Deforestation Exploration -
--
-- =====

-- -----
-- SECTION 1. Database
-- -----
```

```
DROP DATABASE IF EXISTS forestquery;
CREATE DATABASE forestquery;
USE forestquery;
```

```
SELECT DATABASE(); -- should return: forestquery
```

```
-----
-- SECTION 2. Tables
-- Areas are DOUBLE and nullable - the source data has genuine gaps.
-----
```

```
CREATE TABLE forest_area (
  country_code  VARCHAR(10),
  country_name  VARCHAR(100),
  year          INT,
  forest_area_sqkm DOUBLE NULL
);
```

```
CREATE TABLE land_area (
  country_code  VARCHAR(10),
  country_name  VARCHAR(100),
  year          INT,
  total_area_sq_mi DOUBLE NULL
);
```

```
CREATE TABLE regions (
  country_name VARCHAR(100),
  country_code VARCHAR(10),
  region       VARCHAR(100),
  income_group VARCHAR(100)
);
```

```
-----
-- SECTION 3. Enable local file loading
-- Server side. The client side is separate: connection settings ->
-- Advanced -> Others -> OPT_LOCAL_INFILE=1, then reconnect.
-----
```

```
SET GLOBAL local_infile = 1;
SHOW GLOBAL VARIABLES LIKE 'local_infile'; -- Value should read ON
```

```
-----
```

-- SECTION 4. Load the CSVs

```
-----  
LOAD DATA LOCAL INFILE '/Users/wafa/Downloads/forest_area.csv'  
INTO TABLE forest_area  
FIELDS TERMINATED BY ',' OPTIONALLY ENCLOSED BY ''''  
LINES TERMINATED BY '\n'  
IGNORE 1 ROWS  
(country_code, country_name, year, @forest_area_sqkm)  
SET forest_area_sqkm = NULLIF(@forest_area_sqkm, "");
```

```
LOAD DATA LOCAL INFILE '/Users/wafa/Downloads/land_area.csv'  
INTO TABLE land_area  
FIELDS TERMINATED BY ',' OPTIONALLY ENCLOSED BY ''''  
LINES TERMINATED BY '\n'  
IGNORE 1 ROWS  
(country_code, country_name, year, @total_area_sq_mi)  
SET total_area_sq_mi = NULLIF(@total_area_sq_mi, "");
```

SHOW WARNINGS;

```
LOAD DATA LOCAL INFILE '/Users/wafa/Downloads/regions.csv'  
INTO TABLE regions  
FIELDS TERMINATED BY ',' OPTIONALLY ENCLOSED BY ''''  
LINES TERMINATED BY '\n'  
IGNORE 1 ROWS  
(country_name, country_code, region, income_group);
```

SHOW WARNINGS;

-- -----
-- SECTION 5. Sanity checks - do these before trusting any analysis
-- -----

```
SELECT COUNT(*) AS forest_rows FROM forest_area; -- expect ~5900  
SELECT COUNT(*) AS land_rows FROM land_area; -- expect ~5900  
SELECT COUNT(*) AS region_rows FROM regions; -- expect ~220
```

```
SELECT MIN(year) AS first_year, MAX(year) AS last_year FROM forest_area;
```

-- Eyeball a few rows: are the columns in the right places?

```
SELECT * FROM forest_area LIMIT 5;
```

```
SELECT * FROM regions    LIMIT 5;
```

-- How much data is genuinely missing?

```
SELECT SUM(forest_area_sqkm IS NULL) AS missing_forest FROM forest_area;
```

```
SELECT SUM(total_area_sq_mi IS NULL) AS missing_land  FROM land_area;
```

-- The World row must exist: you need it for global figures,

-- and you must EXCLUDE it when ranking regions.

```
SELECT * FROM regions WHERE country_name = 'World';
```

-- SECTION 7. Part 1 deliverable: the forestation view

-- -----

```
CREATE OR REPLACE VIEW forestation AS
```

```
SELECT
```

```
    f.country_code,
```

```
    f.country_name,
```

```
    f.year,
```

```
    f.forest_area_sqkm,
```

```
    l.total_area_sq_mi,
```

```
    l.total_area_sq_mi * 2.59 AS total_area_sqkm,
```

```
    r.region,
```

```
    r.income_group,
```

```
    (f.forest_area_sqkm / (l.total_area_sq_mi * 2.59)) * 100 AS percent_forest
```

```
FROM forest_area AS f
```

```
JOIN land_area AS l
```

```
    ON f.country_code = l.country_code
```

```
    AND f.year        = l.year
```

```
JOIN regions AS r
```

```
    ON f.country_code = r.country_code;
```

```
SELECT * FROM forestation WHERE year = 2016 LIMIT 10;
```

-- SECTION 8. GLOBAL SITUATION

-- -----

```
/*1
```

a. What was the total forest area (in sq km) of the world in 1990?

Please keep in mind that you can use the country record denoted as "World" in the region table.

*/

```
SELECT
SUM(forest_area_sqkm)
FROM forestation
WHERE year = 1990 AND region = 'World';
```

/*

1b. What was the total forest area (in sq km) of the world in 2016?

Please keep in mind that you can use the country record in the table is denoted as "World."

*/

```
SELECT
SUM(forest_area_sqkm)
FROM forestation
WHERE year = 2016 AND region = 'World';
```

/*

1c. What was the change (in sq km) in the forest area of the world from 1990 to 2016?

*/

```
select
(f1.forest_area_sqkm - f2.forest_area_sqkm ) As forest_area_change
from forestation f1
join forestation f2
on
f1.country_name = f2.country_name
Where f1.year ='1990'
AND f2.year = '2016'
AND f1.country_name ='World' ;
```

/*

1d. What was the percent change in forest area of the world between 1990 and 2016?

*/

```
select
((f1.forest_area_sqkm - f2.forest_area_sqkm )) *100/ (f1.forest_area_sqkm )As
forest_area_precentage
from forestation f1
join forestation f2
on
f1.country_name = f2.country_name
Where f1.year ='1990'
AND f2.year = '2016'
AND f1.country_name ='World';
```

/*

1e. If you compare the amount of forest area lost between 1990 and 2016,
to which country's total area in 2016 is it closest to?

*/

```
WITH loss AS (  
    SELECT MAX(CASE WHEN year = 1990 THEN forest_area_sqkm END)  
        - MAX(CASE WHEN year = 2016 THEN forest_area_sqkm END) AS forest_lost_sqkm  
    FROM forestation  
    WHERE country_name = 'World'  
)  
SELECT f.country_name,  
    ROUND(f.total_area_sq_mi * 2.59, 2) AS total_area_sqkm,  
    ROUND(ABS(f.total_area_sq_mi * 2.59 - l.forest_lost_sqkm), 2) AS diff_sqkm  
FROM forestation f  
CROSS JOIN loss l  
WHERE f.year = 2016  
    AND f.total_area_sq_mi IS NOT NULL  
    AND f.region IS NOT NULL  
    AND f.country_name != 'World'  
ORDER BY diff_sqkm ASC  
LIMIT 1;  
-- SECTION 8. REGIONAL OUTLOOK
```

/*2a. What was the percent forest of the entire world in 2016?

Which region had the HIGHEST percent forest in 2016,
and which had the LOWEST, to 2 decimal places?

*/

-- percent--

```
SELECT forest_area_sqkm * 100 / total_area_sqkm  
FROM forestation  
WHERE year = 2016  
AND country_name = 'World';
```

-- heighest--

```
SELECT region,  
    ROUND(percent_forest , 2)  
FROM  
(SELECT region,  
    SUM(forest_area_sqkm)*100/SUM(total_area_sqkm) AS percent_forest  
FROM forestation  
WHERE year = 2016 and region != 'World'  
GROUP BY 1 ) sub
```

```

ORDER BY 2 DESC
LIMIT 1;
-- lowest--
SELECT region,
ROUND(percent_forest , 2)
FROM
(SELECT region,
SUM(forest_area_sqkm)*100/SUM(total_area_sqkm) AS percent_forest
FROM forestation
WHERE year = 2016 and region != 'World'
GROUP BY 1 ) sub
ORDER BY 2 ASC
LIMIT 1;

```

/*

2b. What was the percent forest of the entire world in 1990?

Which region had the HIGHEST percent forest in 1990,
and which had the LOWEST, to 2 decimal places?

*/

-- precent--

```

SELECT forest_area_sqkm * 100 / total_area_sqkm
FROM forestation
WHERE year = 1990
AND country_name = 'World';

```

-- heighest--

```

SELECT region,
ROUND(percent_forest , 2)
FROM
(SELECT region,
SUM(forest_area_sqkm)*100/SUM(total_area_sqkm) AS percent_forest
FROM forestation
WHERE year = 1990 and region != 'World'
GROUP BY 1 ) sub
ORDER BY 2 DESC
LIMIT 1;

```

-- lowest--

```

SELECT region,
ROUND(percent_forest , 2)
FROM
(SELECT region,
SUM(forest_area_sqkm)*100/SUM(total_area_sqkm) AS percent_forest
FROM forestation

```

```
WHERE year = 1990 and region != 'World'
GROUP BY 1 ) sub
ORDER BY 2 ASC
LIMIT 1;
```

/*2c. Based on the table you created, which regions of the world
DECREASED in forest area from 1990 to 2016?
*/

```
WITH tb1 AS(SELECT
region,
SUM(forest_area_sqkm) AS region_forest_sum_1990
FROM forestation
WHERE year = 1990 AND region != 'World'
GROUP BY region),
tb2 AS(SELECT
region,
SUM(forest_area_sqkm) AS region_forest_sum_2016
FROM forestation
WHERE year = 2016 AND region != 'World'
GROUP BY region)
SELECT
tb1.region,
region_forest_sum_1990,
region_forest_sum_2016
from tb1
join tb2
on tb1.region = tb2.region
where region_forest_sum_1990 > region_forest_sum_2016
;
```

-- Table --

```
WITH tb1 AS(
SELECT
region,
SUM(forest_area_sqkm) AS region_forest_sum_1990,
SUM(total_area_sq_mi * 2.59) AS region_land_sum_1990
FROM forestation
WHERE year = 1990
GROUP BY region),
tb2 AS(SELECT
region,
```

```

SUM(forest_area_sqkm) AS region_forest_sum_2016,
SUM(total_area_sq_mi * 2.59) AS region_land_sum_2016
FROM forestation
WHERE year = 2016
GROUP BY region)
SELECT
tb1.region,
ROUND(region_forest_sum_1990 * 100.0 / region_land_sum_1990, 2) AS `1990 Forest
Percentage`,
ROUND(region_forest_sum_2016 * 100.0 / region_land_sum_2016, 2) AS `2016 Forest
Percentage`
from tb1
join tb2
on tb1.region = tb2.region
order by `1990 Forest Percentage` desc
;
-- SECTION 9. Country-Level Detail

```

/*

Which 5 countries saw the largest amount decrease in forest area from 1990 to 2016? What was the difference in forest area for each?

*/

```

WITH tb1 AS (
  SELECT country_name, region, forest_area_sqkm AS forest_1990
  FROM forestation
  WHERE year = 1990 AND region!= 'World'),
tb2 AS (
  SELECT country_name, forest_area_sqkm AS forest_2016
  FROM forestation
  WHERE year = 2016 AND region!= 'World')
SELECT DISTINCT tb1.country_name,
  tb1.region,
  ROUND(forest_1990 - forest_2016, 2) AS Absolute_Forest_Area_Change,
  ROUND(((forest_1990 - forest_2016)*100)/forest_1990, 2) AS decrease_precentage
FROM tb1
JOIN tb2 ON tb1.country_name = tb2.country_name
ORDER BY Absolute_Forest_Area_Change DESC
LIMIT 5;

```

/*

Which 5 countries saw the largest percent decrease in forest area from 1990 to 2016? What was the percent change to 2 decimal places for each?

*/

```
WITH tb1 AS (  
    SELECT country_name, region, forest_area_sqkm AS forest_1990  
    FROM forestation  
    WHERE year = 1990 AND region!= 'World'),  
tb2 AS (  
    SELECT country_name, forest_area_sqkm AS forest_2016  
    FROM forestation  
    WHERE year = 2016 AND region!= 'World')  
SELECT distinct tb1.country_name,  
    tb1.region,  
    ROUND(forest_1990 - forest_2016, 2) AS forest_loss_sqkm,  
    ROUND(((forest_1990 - forest_2016)*100)/forest_1990, 2) AS decrease_precentage  
FROM tb1  
JOIN tb2 ON tb1.country_name = tb2.country_name  
ORDER BY decrease_precentage DESC  
LIMIT 5;
```

/*

SUCCESS STORIES

*/

```
WITH tb1 AS(SELECT  
country_code,  
country_name,  
region,  
forest_area_sqkm AS forest_1990  
FROM forestation  
WHERE year = 1990 AND region !='World'),  
tb2 AS (SELECT  
country_code,  
forest_area_sqkm AS forest_2016  
FROM forestation  
WHERE year = 2016 AND region !='World')  
SELECT distinct  
tb1.country_name,  
ROUND(forest_1990,2) AS forest_1990,  
ROUND(forest_2016,2) AS forest_2016,  
ROUND(forest_2016 - forest_1990, 2) AS change_sqkm,  
ROUND((((forest_2016 - forest_1990) / forest_1990) * 100, 2) AS pct_change  
from tb1
```

```

join tb2
on tb1.country_code = tb2.country_code
where forest_1990 < forest_2016
order by change_sqkm desc
limit 5
;

```

```

/*
3b. Country with largest percent change in forest area from 1990 to 2016
*/

```

```

With tb1 AS
(SELECT country_name, year, forest_area_sqkm, total_area_sq_mi*2.59
  AS total_area_sqkm, percent_forest
FROM forestation
WHERE (year='2016' AND region != 'World'
      AND forest_area_sqkm != 0 AND total_area_sq_mi != 0)
ORDER BY percent_forest DESC),

```

```

tb2 AS
(SELECT tb1.country_name, tb1.year, tb1.percent_forest,
  CASE WHEN tb1.percent_forest > 75 THEN 4
  WHEN tb1.percent_forest <= 75 AND tb1.percent_forest > 50 THEN 3
  WHEN tb1.percent_forest <= 50 AND tb1.percent_forest > 25 THEN 2
  ELSE 1
  END AS percentile
FROM tb1
ORDER BY 4 DESC)

```

```

SELECT
tb2.percentile, COUNT(tb2.percentile) AS act
FROM tb2
GROUP BY tb2.percentile
ORDER BY act DESC;

```

```

/*
3c. If countries were grouped by percent forestation in quartiles,
which group had the most countries in it in 2016?
*/

```

```

SELECT country_name, region, year, forest_area_sqkm, total_area_sq_mi*2.59 AS
total_area_sqkm,
ROUND(percent_forest, 2) AS percent
FROM forestation

```

```
WHERE (year='2016' AND region!='World'
      AND forest_area_sqkm !=0 AND total_area_sq_mi!=0)
      AND percent_forest > 75
```

```
ORDER BY percent_forest DESC
```

```
limit 10;
```

```
/*
```

3e.How many countries had a percent forestation higher than the United States in 2016?

```
*/
```

With tb1 AS

```
(SELECT country_name, year,forest_area_sqkm, total_area_sq_mi*2.59
```

```
  AS total_area_sqkm, percent_forest
```

```
  FROM forestation
```

```
  WHERE (year='2016' AND region!='World'
```

```
        AND forest_area_sqkm !=0 AND total_area_sq_mi!=0)
```

```
  ORDER BY percent_forest DESC)
```

```
SELECT COUNT(tb1.country_name)
```

```
FROM tb1
```

```
WHERE tb1.percent_forest > (SELECT tb1.percent_forest
```

```
  FROM tb1
```

```
  WHERE tb1.country_name = 'United States');
```